

Student Exam Results – PCA Analysis

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Descripción

La base **Student_Exam_Results.csv** contiene calificaciones de exámenes de 1000 estudiantes en seis asignaturas.

Incluye tres columnas adicionales:

- **Total** (suma de todas las calificaciones, debe eliminarse)
 - **Results**: 0 = no aprobado, 1 = aprobado
 - **Div**: categorías 0, 1, 2, 3 (significado no claro)
-

Librerías

```
#library(learnr)
library(tidyverse)
library(tidymodels)
library(embed)
library(ggcorrplot)
library(corr)
library(tidytext)
#library(gradethis)
library(psych)
#library(sortable)
#library(learntidymodels)
library(rstatix)
library(broom)
library(rgl)
library(plotly)
library(GGally)
library(tidyr)
library(FactoMineR)
library(factoextra)
library(ggord)

theme_set(theme_bw(16))
```

Importación de Datos

```
df <- read.csv("C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Student_Results/Student_Exam_Resu~  
dim(df)
```

```
## [1] 1000    9
```

```
glimpse(df)
```

```
## Rows: 1,000  
## Columns: 9  
## $ Hindi      <int> 65, 94, 7, 88, 39, 51, 73, 18, 15, 53, 61, 49, 21, 93, 44, 5~  
## $ English    <int> 10, 56, 85, 46, 81, 43, 48, 23, 64, 85, 17, 10, 24, 57, 6, 5~  
## $ Science    <int> 59, 4, 76, 59, 37, 14, 6, 97, 14, 96, 16, 81, 83, 6, 72, 39,~  
## $ Maths      <int> 3, 67, 99, 94, 38, 53, 38, 65, 43, 76, 16, 13, 93, 94, 59, 9~  
## $ History    <int> 71, 91, 60, 52, 6, 64, 50, 15, 59, 83, 20, 39, 93, 23, 40, 5~  
## $ Geograpgy  <int> 37, 50, 25, 38, 54, 59, 21, 20, 59, 62, 48, 86, 28, 84, 6, 4~  
## $ Total      <int> 245, 362, 352, 377, 255, 284, 236, 238, 254, 455, 178, 278, ~  
## $ Results    <int> 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, ~  
## $ Div        <int> 3, 1, 2, 1, 3, 2, 3, 3, 3, 1, 0, 2, 2, 2, 3, 2, 3, 2, 2, 2, ~
```

```
head(df)
```

```
##   Hindi English Science Maths History Geograpgy Total Results Div  
## 1    65     10      59     3      71        37    245        0    3  
## 2    94     56       4     67     91        50    362        1    1  
## 3     7     85      76    99     60        25    352        0    2  
## 4    88     46      59    94     52        38    377        1    1  
## 5    39     81      37    38      6        54    255        1    3  
## 6    51     43      14    53     64        59    284        1    2
```

Preparación de Datos

Selección de columnas relevantes

```
df1 <- df[,1:9]  
head(df1)
```

```
##   Hindi English Science Maths History Geograpgy Total Results Div  
## 1    65     10      59     3      71        37    245        0    3  
## 2    94     56       4     67     91        50    362        1    1  
## 3     7     85      76    99     60        25    352        0    2  
## 4    88     46      59    94     52        38    377        1    1  
## 5    39     81      37    38      6        54    255        1    3  
## 6    51     43      14    53     64        59    284        1    2
```

Valores faltantes

```
colSums(is.na(df1))
```

```
##      Hindi   English   Science   Maths   History Geograpgy   Total   Results
##         0         0         0         0         0         0         0         0
##      Div
##         0
```

```
summary(df1)
```

```
##      Hindi      English      Science      Maths
## Min.   : 0.00   Min.   : 0.00   Min.   : 0.00   Min.   : 0.00
## 1st Qu.:26.00   1st Qu.:26.00   1st Qu.:25.00   1st Qu.:25.75
## Median :53.00   Median :50.50   Median :50.00   Median :49.00
## Mean   :51.65   Mean   :50.11   Mean   :49.44   Mean   :49.55
## 3rd Qu.:77.00   3rd Qu.:75.00   3rd Qu.:73.25   3rd Qu.:74.00
## Max.   :99.00   Max.   :99.00   Max.   :99.00   Max.   :99.00
##      History      Geograpgy      Total      Results      Div
## Min.   : 0.00   Min.   : 0.00   Min.   :103.0   Min.   :0.00   Min.   :0.000
## 1st Qu.:24.00   1st Qu.:26.00   1st Qu.:254.0   1st Qu.:0.00   1st Qu.:1.000
## Median :49.00   Median :49.00   Median :296.0   Median :0.00   Median :2.000
## Mean   :49.03   Mean   :50.03   Mean   :299.8   Mean   :0.35   Mean   :1.856
## 3rd Qu.:73.25   3rd Qu.:75.00   3rd Qu.:349.2   3rd Qu.:1.00   3rd Qu.:3.000
## Max.   :99.00   Max.   :99.00   Max.   :505.0   Max.   :1.00   Max.   :3.000
```

Eliminación de la columna Total

```
df1 <- df1 %>% select(-Total)
```

Conversión a factores

```
df1 <- df1 %>%
  mutate(
    Results = as.factor(Results),
    Div = as.factor(Div)
  )
count(df1, Results)
```

```
## Results    n
## 1         0 650
## 2         1 350
```

```
count(df1, Div)
```

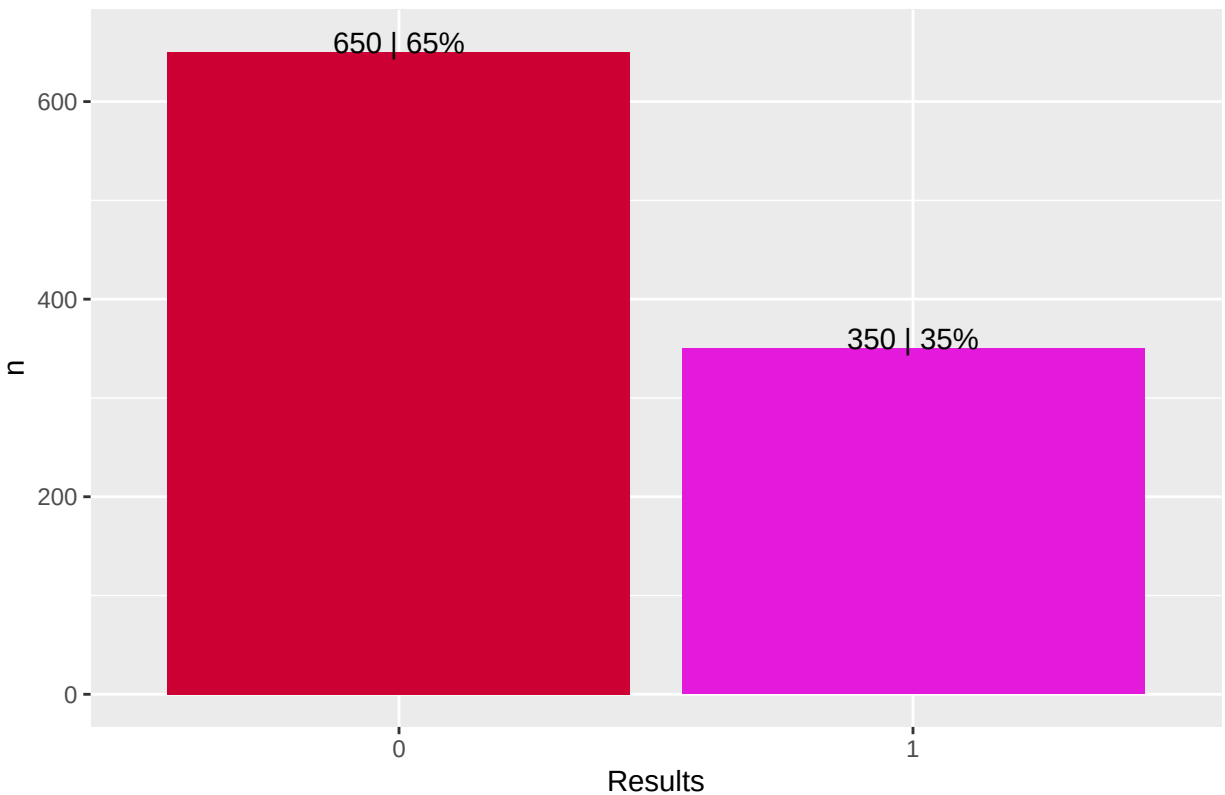
```
##   Div   n
## 1    0  95
## 2    1 206
## 3    2 447
## 4    3 252
```

Distribuciones de variables objetivo

Distribución de Results

```
df1 %>%
  group_by(Results) %>%
  summarise(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = Results, y = n)) +
  geom_col(fill = c("#CC0033", "#e319dc")) +
  geom_text(
    aes(label = paste0(n, " | ", signif(prop * 100, digits = 4), "%")),
    nudge_y = 10
  ) +
  ggtitle("Distribución de Resultados (No Aprobado vs Aprobado)") +
  theme_gray()
```

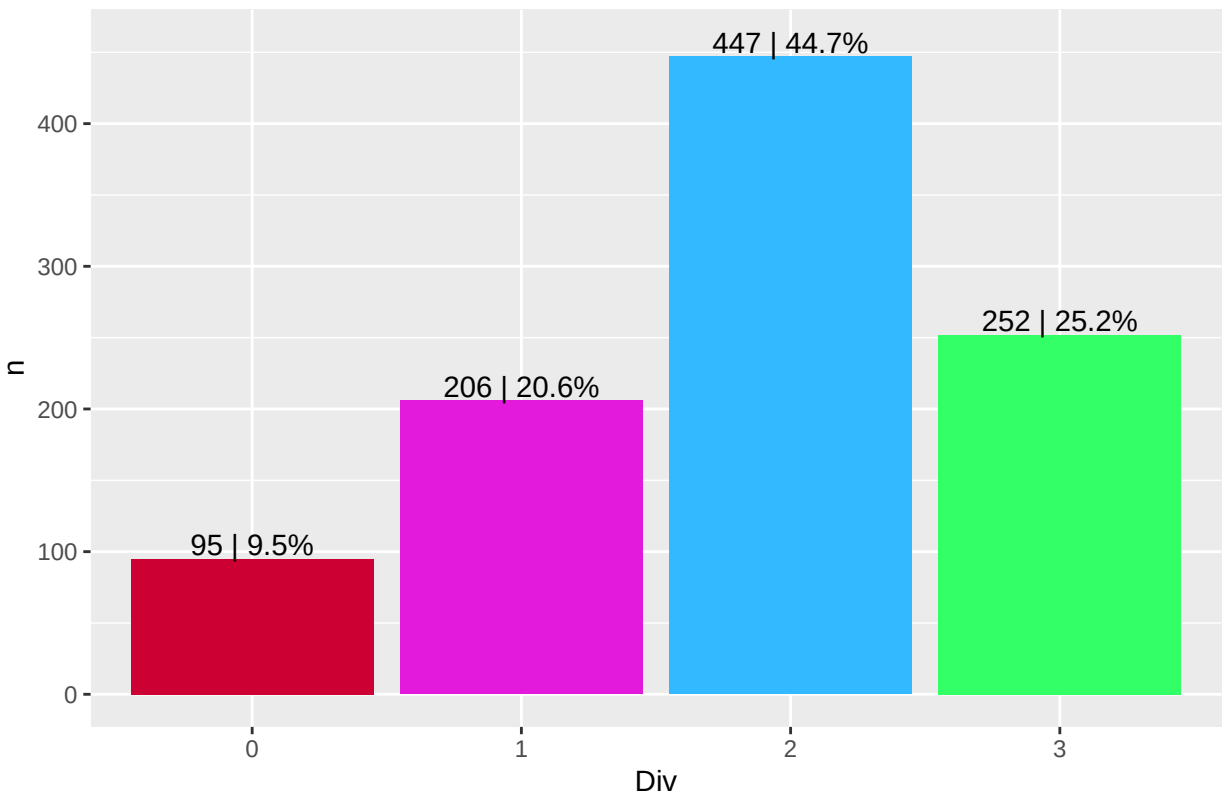
Distribución de Resultados (No Aprobado vs Aprobado)



Distribución de Div

```
df1 %>%
  group_by(Div) %>%
  summarise(n = n()) %>%
  mutate(prop = n / sum(n)) %>%
  ggplot(aes(x = Div, y = n)) +
  geom_col(fill = c("#CC0033", "#e319dc", "#33b9ff", "#33ff66")) +
  geom_text(
    aes(label = paste0(n, " | ", signif(prop * 100, digits = 4), "%")),
    nudge_y = 10
  ) +
  ggtitle("Distribución de Divisiones (0, 1, 2, 3)") +
  theme_gray()
```

Distribución de Divisiones (0, 1, 2, 3)



Análisis Exploratorio

Medias de variables numéricas

```
df1 %>%
  select(where(is.numeric)) %>%
  colMeans()
```

```
##      Hindi   English   Science   Maths   History Geograpgy
##    51.645    50.110    49.440    49.553    49.034    50.028
```

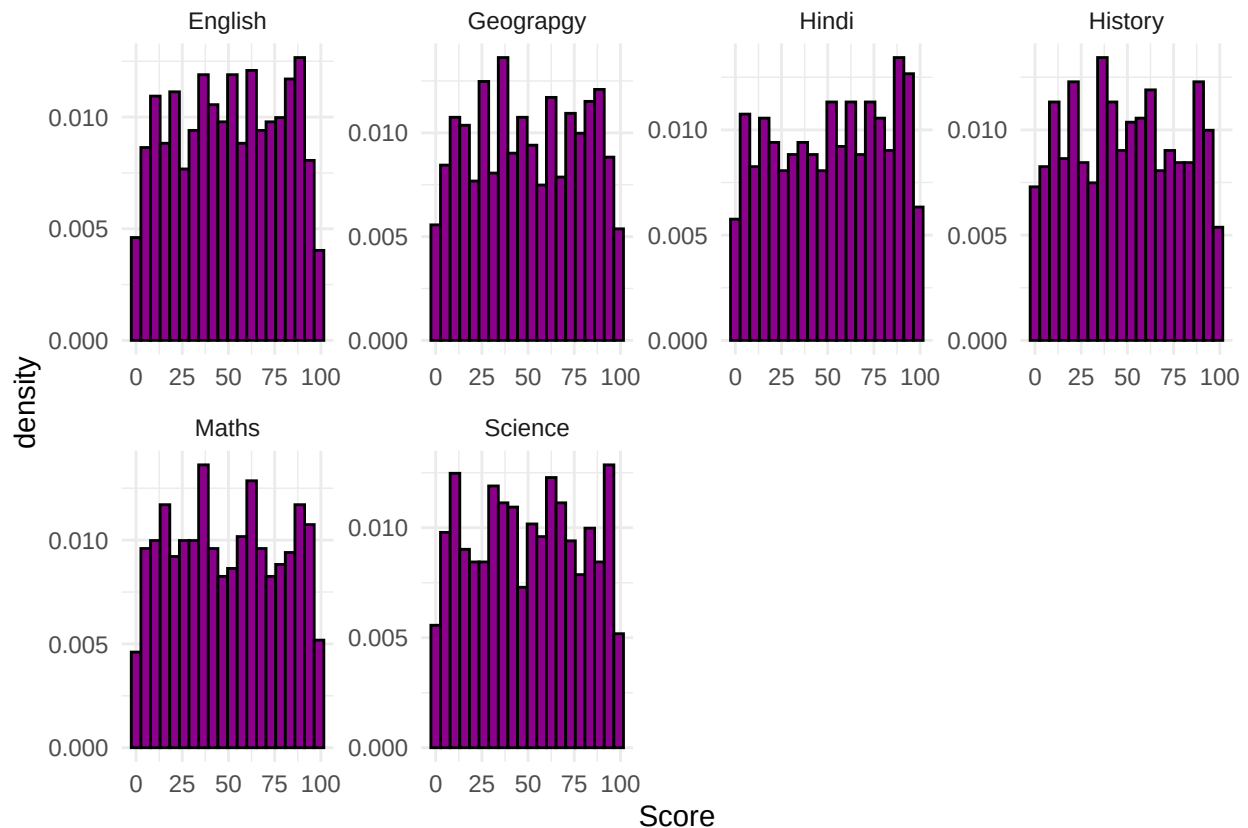
Histogramas por variable

```
df1 %>%
  pivot_longer(
```

```

cols = !c(Results, Div),
names_to = "Variable",
values_to = "Score"
) %>%
ggplot(aes(x = Score)) +
geom_histogram(aes(y = ..density..), bins = 20,
               colour = "black", fill = "darkmagenta") +
facet_wrap(~ Variable, ncol = 4, scales = "free") +
theme_minimal()

```



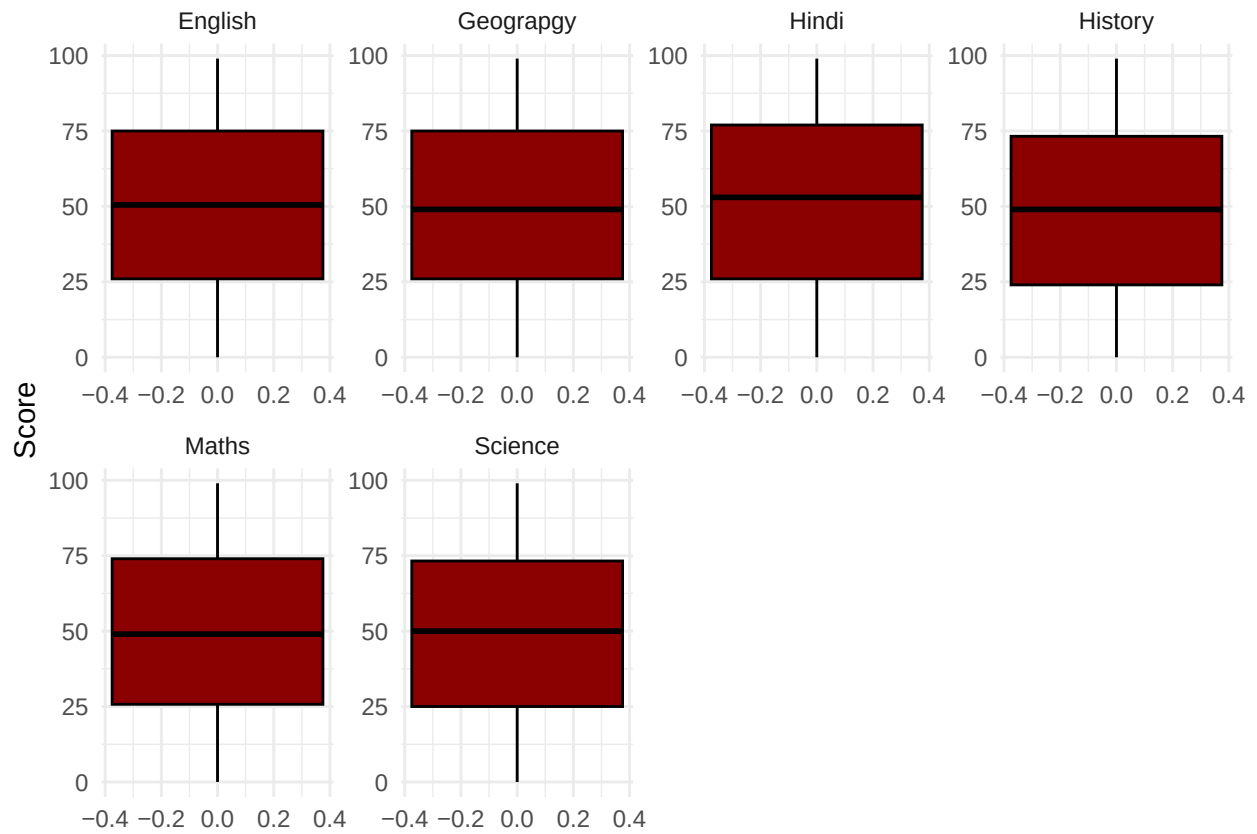
Boxplots por variable

```

df1 %>%
pivot_longer(
  cols = !c(Results, Div),
  names_to = "Variable",
  values_to = "Score"
) %>%
ggplot(aes(y = Score)) +
geom_boxplot(fill = "darkred", color = "black") +

```

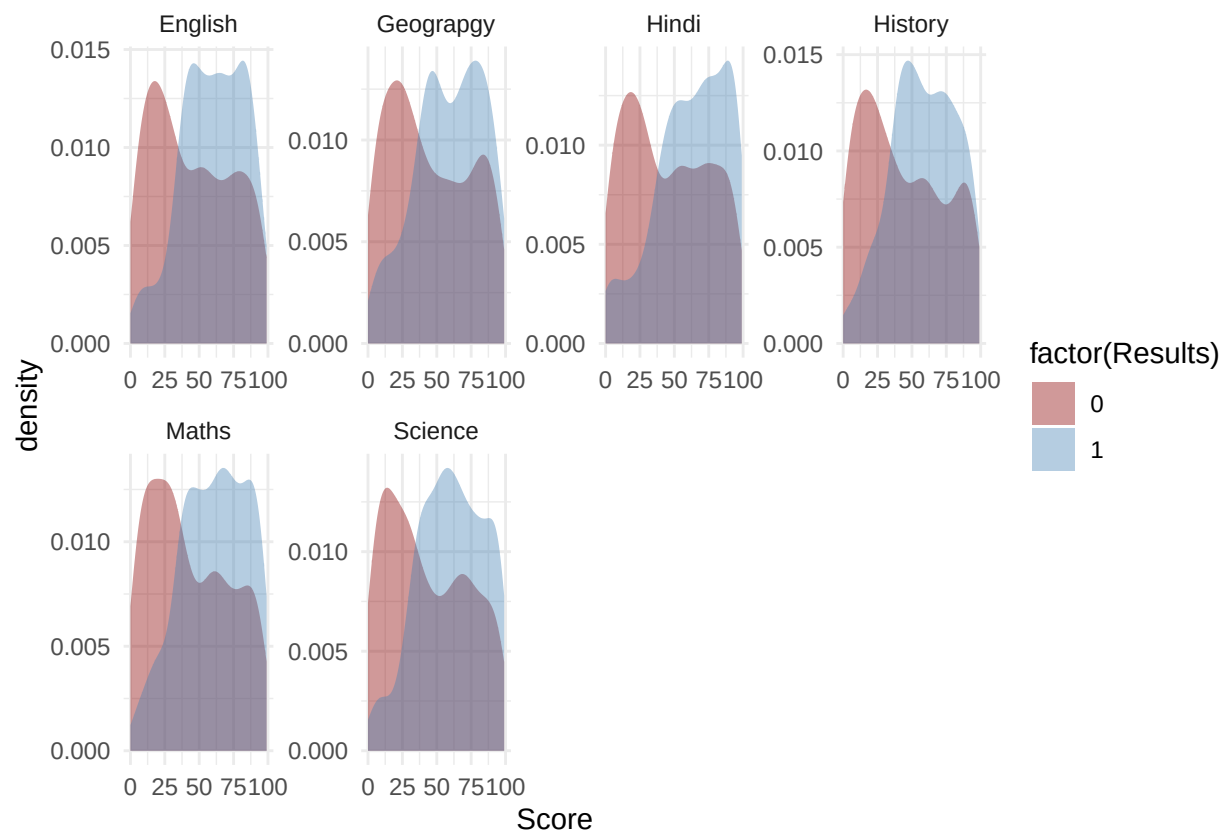
```
facet_wrap(~ Variable, ncol = 4, scales = "free") +
theme_minimal()
```



Gráficas de densidad

Por Results

```
df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score, fill = factor(Results))) +
  geom_density(alpha = 0.4, colour = NA) +
  facet_wrap(~ Variable, ncol = 4, scales = "free") +
  scale_fill_manual(values = c("0"="darkred", "1"="steelblue")) +
  theme_minimal()
```

Por Div

```
df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score, fill = factor(Div))) +
  geom_density(alpha = 0.4, colour = NA) +
  facet_wrap(~ Variable, ncol = 4, scales = "free") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```



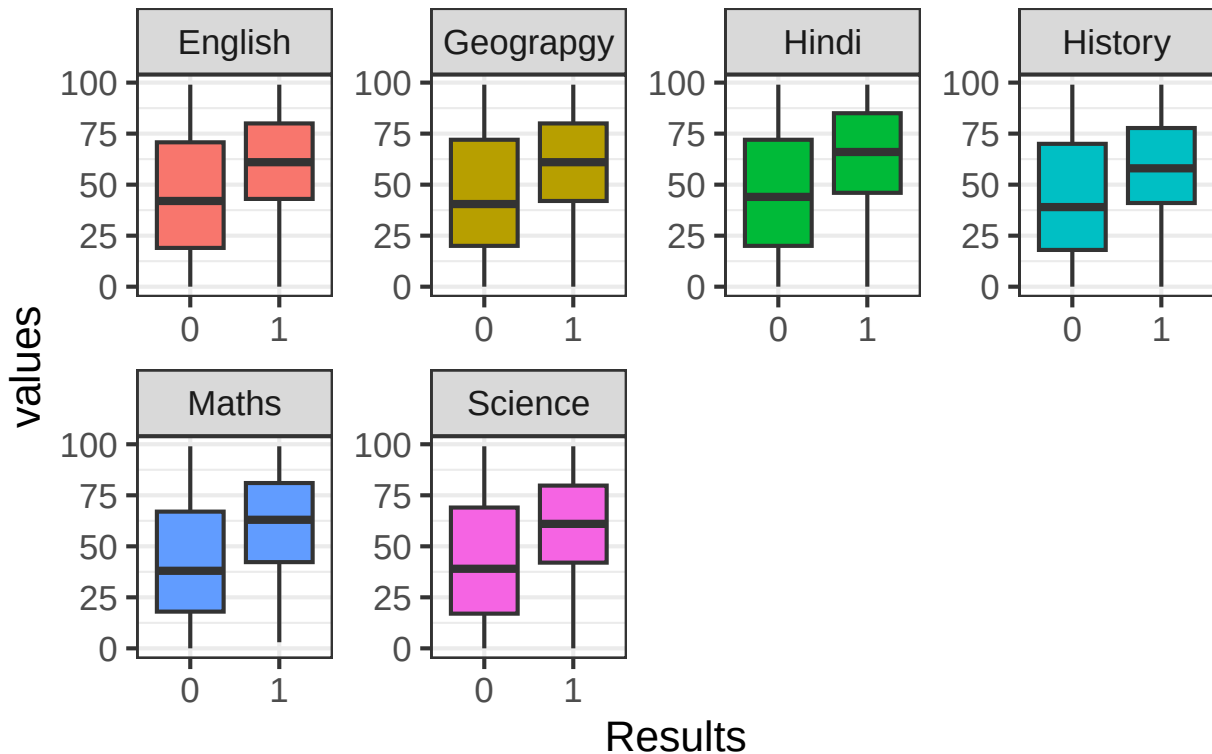
Comparación por variable de respuesta (Results)

```
df1_long <- df1 %>%
  pivot_longer(cols = !c(Results, Div), names_to = "predictores", values_to = "values")
```

Boxplots

```
df1_long %>%
  ggplot(aes(x = Results, y = values, fill = predictores)) +
  geom_boxplot() +
  facet_wrap(~ predictores, scales = "free", ncol = 4) +
  theme(legend.position = "none") +
  labs(title = "Comparación vía box-plot")
```

Comparación vía box-plot

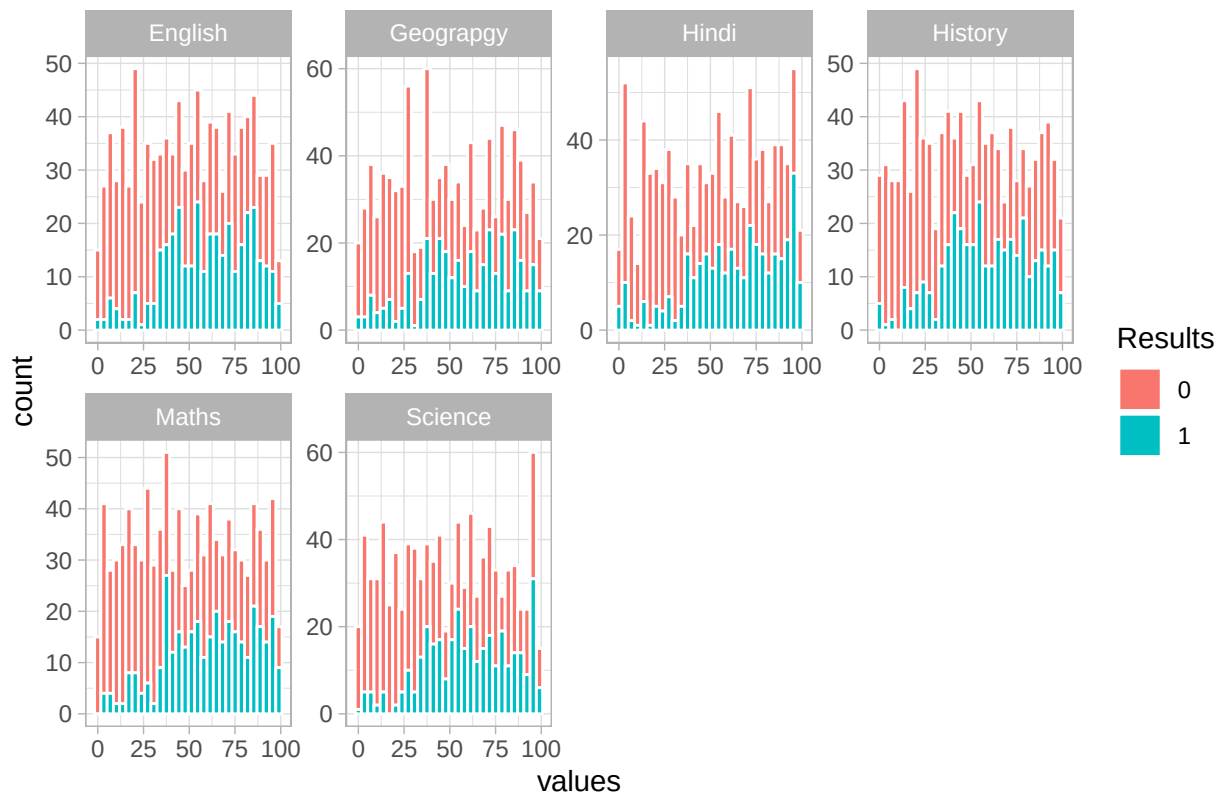


Histogramas

```
df1_long %>%
  ggplot(aes(values, fill = Results)) +
  geom_histogram(color = "white") +
  facet_wrap(~ predictores, scales = "free", ncol=4) +
  theme_light() +
  labs(title = "Comparación vía histograma")
```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

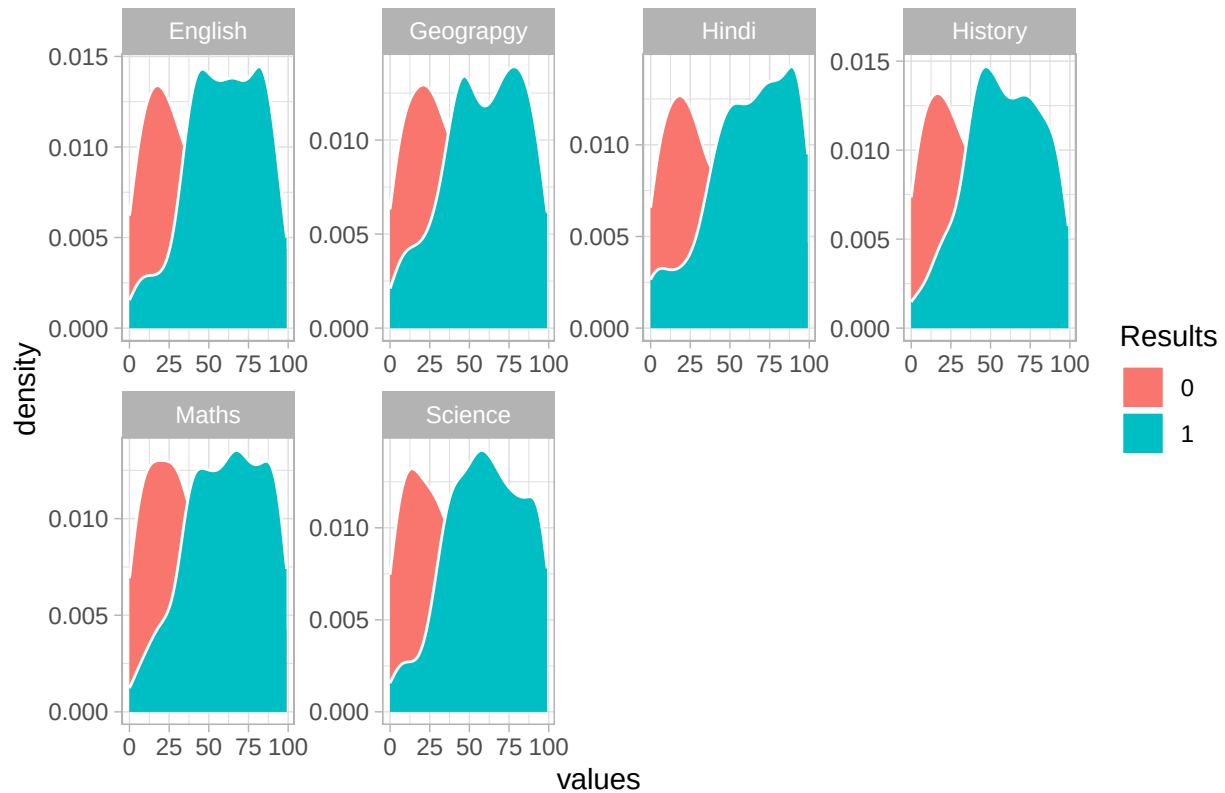
Comparación vía histograma



Densidad

```
df1_long %>%
  ggplot(aes(values, fill = Results)) +
  geom_density(color = "white") +
  facet_wrap(~ predictores, scales = "free", ncol=4) +
  theme_light() +
  labs(title = "Comparación a través de densidad")
```

Comparación a través de densidad



Matriz de correlación

```
grades_corr <- df1 %>%
  select(-Results, -Div) %>%
  correlate(method = "spearman")
```

```
## Correlation computed with
## * Method: 'spearman'
## * Missing treated using: 'pairwise.complete.obs'
```

```
grades_corr
```

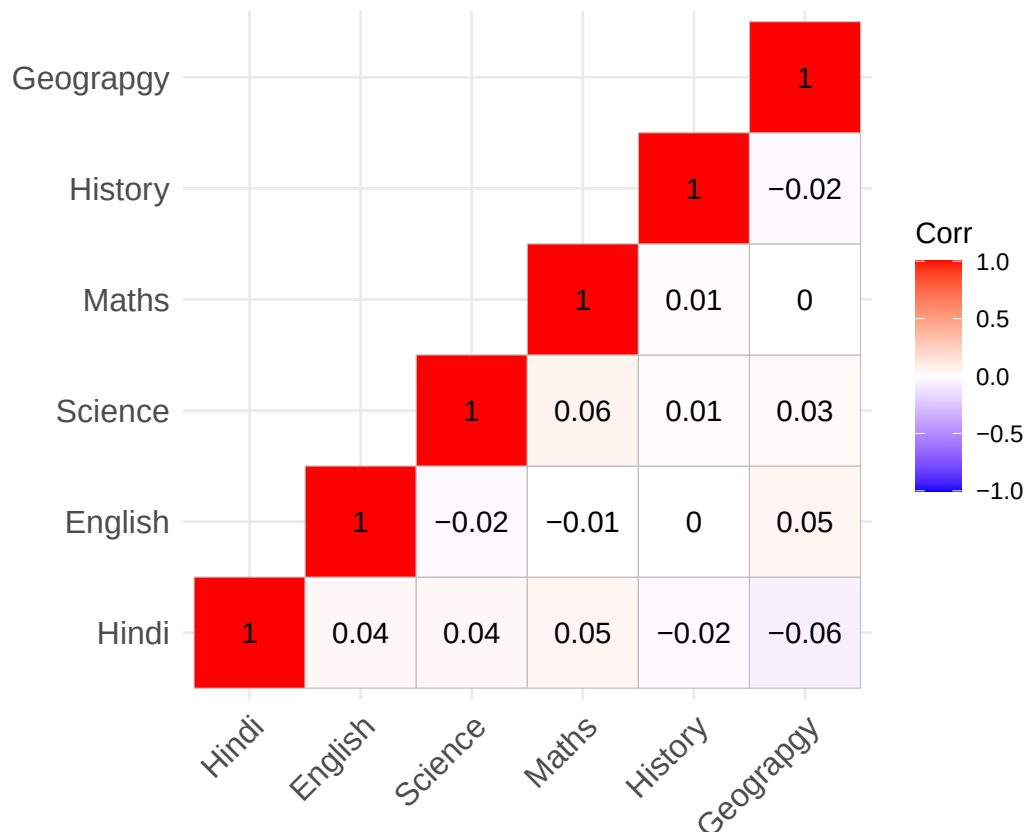
```
## # A tibble: 6 x 7
##   term      Hindi English Science Maths History Geography
##   <chr>    <dbl>   <dbl>   <dbl> <dbl>   <dbl>    <dbl>
## 1 Hindi    NA      0.0370  0.0387  0.0454 -0.0212  -0.0633
## 2 English  0.0370 NA      -0.0234 -0.0131 -0.00114  0.0523
## 3 Science  0.0387 -0.0234 NA      0.0559  0.00565  0.0285
## 4 Maths    0.0454 -0.0131  0.0559 NA      0.0121  -0.00115
## 5 History  -0.0212 -0.00114  0.00565  0.0121 NA      -0.0209
## 6 Geography -0.0633  0.0523  0.0285 -0.00115 -0.0209 NA
```

Correlation matrix (base R)

```
corr_stg <- cor(df1[, -c(7, 8)], method = "spearman")
corr_stg
```

```
##           Hindi      English      Science      Maths      History
## Hindi      1.00000000  0.036962705  0.038719755  0.045364412 -0.021182424
## English    0.03696271  1.000000000 -0.023392589 -0.013091984 -0.001138727
## Science    0.03871975 -0.023392589  1.000000000  0.055931481  0.005645743
## Maths      0.04536441 -0.013091984  0.055931481  1.000000000  0.012092849
## History    -0.02118242 -0.001138727  0.005645743  0.012092849  1.000000000
## Geograpgy -0.06325747  0.052349020  0.028464649 -0.001147157 -0.020918686
##           Geograpgy
## Hindi      -0.063257467
## English     0.052349020
## Science     0.028464649
## Maths      -0.001147157
## History     -0.020918686
## Geograpgy  1.000000000
```

```
ggcorrplot::ggcorrplot(corr = corr_stg,
                        type = "lower",
                        show.diag = TRUE,
                        lab = TRUE)
```



Determinante

```
det(corr_stg)
```

```
## [1] 0.9822475
```

KMO

```
psych::KMO(corr_stg)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: psych::KMO(r = corr_stg)
## Overall MSA = 0.48
## MSA for each item =
##      Hindi   English   Science   Maths   History Geograpgy
##      0.48     0.46     0.49     0.52     0.46     0.47
```

Test de Bartlett

```
psych::cortest.bartlett(corr_stg, n = dim(df1)[1])
```

```
## $chisq
## [1] 17.84333
##
## $p.value
## [1] 0.2709919
##
## $df
## [1] 15
```

Preparación para PCA

```
pca_recipe <- recipe(~., data = df1) %>%
  update_role(Results, new_role = "id") %>%
  update_role(Div, new_role = "id") %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_predictors()) %>%
  step_pca(all_numeric_predictors(), threshold = 0.80, id="pca")

pca_recipe
```

```
##
```

```
## -- Recipe -----
##
## -- Inputs

## Number of variables by role

## predictor: 6
## id:         2

##

## -- Operations

## * Dummy variables from: all_nominal_predictors()

## * Centering and scaling for: all_predictors()

## * PCA extraction with: all_numeric_predictors()
```

Ejecución del PCA

```
stg_pca <- prep(pca_recipe)
```

Loadings

```
pca_loading <- tidy(stg_pca, id="pca")
pca_loading
```

```
## # A tibble: 36 x 4
##   terms      value component id
##   <chr>      <dbl> <chr>   <chr>
## 1 Hindi    -0.583    PC1     pca
## 2 English   0.133    PC1     pca
## 3 Science  -0.482    PC1     pca
## 4 Maths    -0.551    PC1     pca
## 5 History  -0.0228   PC1     pca
## 6 Geograpgy 0.326     PC1     pca
## 7 Hindi     0.000405 PC2     pca
## 8 English  -0.483    PC2     pca
## 9 Science  -0.380    PC2     pca
## 10 Maths   -0.223    PC2     pca
## # i 26 more rows
```


Varianzas

```
pca_variances <- tidy(stg_pca, id="pca", type="variance")
pca_variances
```

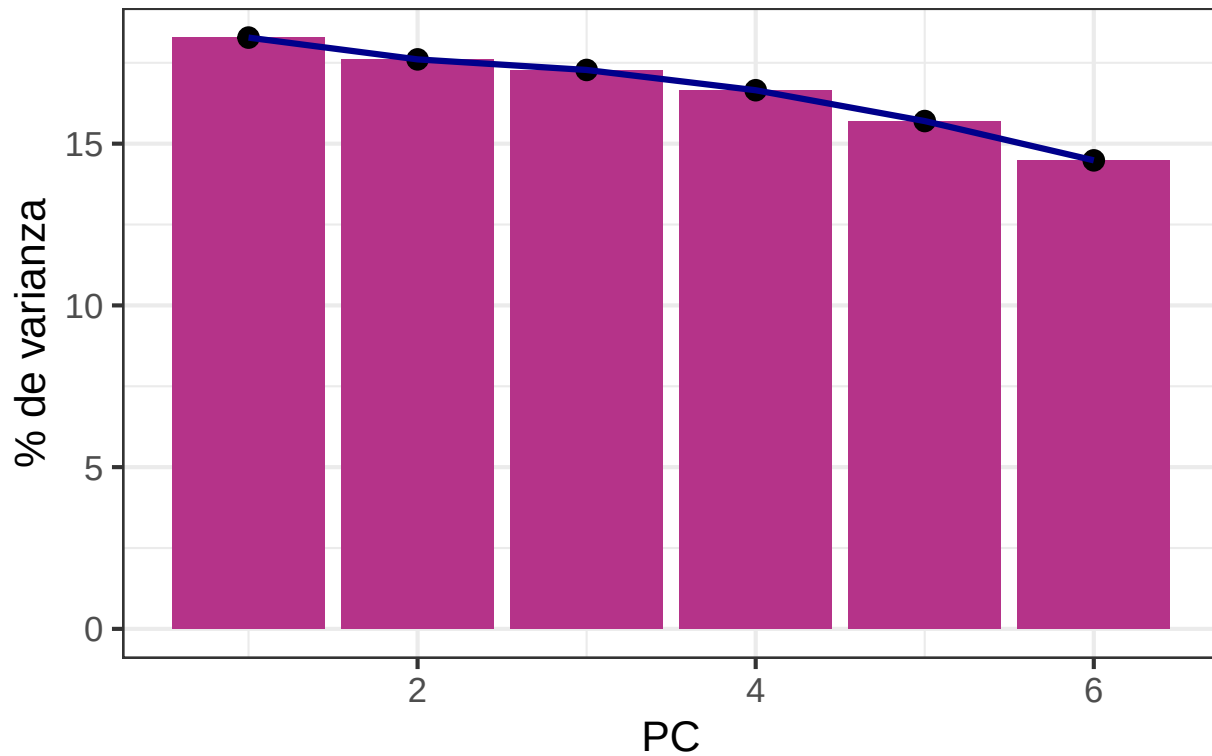
```
## # A tibble: 24 x 4
##   terms          value component id
##   <chr>         <dbl>     <int> <chr>
## 1 variance      1.10         1  pca
## 2 variance      1.06         2  pca
## 3 variance      1.04         3  pca
## 4 variance      0.999        4  pca
## 5 variance      0.942        5  pca
## 6 variance      0.869        6  pca
## 7 cumulative variance 1.10         1  pca
## 8 cumulative variance 2.15         2  pca
## 9 cumulative variance 3.19         3  pca
## 10 cumulative variance 4.19         4  pca
## # i 14 more rows
```

Scree plots

% de varianza por componente

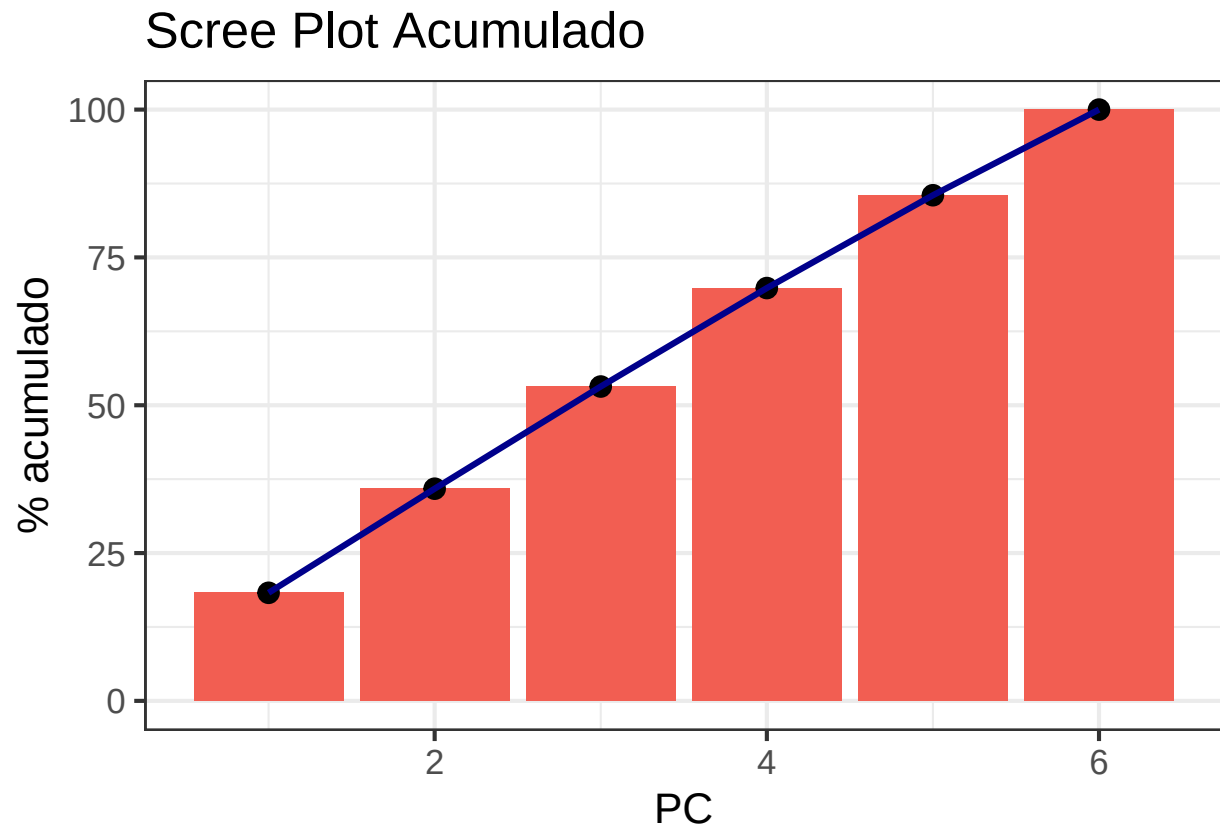
```
stg_pca %>%
  tidy(id="pca", type="variance") %>%
  filter(terms == "percent variance") %>%
  ggplot(aes(x = component, y = value)) +
  geom_col(fill="#B53389") +
  geom_point(size=3) +
  geom_line(color="darkblue", size=1.1) +
  labs(x="PC", y="% de varianza", title="Scree Plot")
```

Scree Plot



% acumulado de varianza

```
stg_pca %>%  
  tidy(id="pca", type="variance") %>%  
  filter(terms == "cumulative percent variance") %>%  
  ggplot(aes(x = component, y = value)) +  
  geom_col(fill="#F25E52") +  
  geom_point(size=3) +  
  geom_line(color="darkblue", size=1.1) +  
  labs(x="PC", y="% acumulado", title="Scree Plot Acumulado")
```



Conclusiones

- Solo se retiene alrededor del **85%** de la información con 5 componentes.
 - La correlación entre variables es **muy baja**, por lo que PCA no es la mejor técnica, pero se realizó por motivos de práctica.
-